

Osiris

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Database Design and Requirements

Functional Requirements

- Users must be able to register and login.
- Users must be able to search for pets based on filters in pet attributes.
- The system must be able to support applications being sent.
- The system should be capable of removing pets from the search function if unavailable.
- U.I. must advertise the animals in an appealing way.

Non-Functional Requirements

- User data must be encrypted.
- When a pet is adopted the system should be able to automatically reject all other applications.
- Applicants must be noticed if an application is accepted and asked if they would like to keep any other applications open
- Applications must move through the following process: Submitted → Reviewed → Approved/Rejected
- Adoption approval and *pet* status change must occur in the same transaction.

Business Rules

- Each *pet* can only be adopted once.
- Each *pet* must have at least 1 *medical record*.
- Each *adoption* record must be linked to one approved *application*.
- Every *applicant* must have a unique email and conform to a valid email format.
- *Adoption* fees must be ≥ 0 and ≤ 100 .
- *Adoption* dates must be on/after the application date and pet's intake date.
- If the *medical record* is a vaccination, a vaccine type must be defined.

Database Entities

1. Applicant

Attribute	Key	Description
ApplicantID	PK	Unique ID for each applicant
FirstName		Applicant's first name
LastName		Applicant's last name
Email		Applicant's unique email address
Phone		Applicant's phone number
Address		Applicant's full address

2. Pet

Attribute	Key	Description
PetID	PK	Unique pet identifier
Name		Pet's name
PetType		Species/type (Dog, Cat, etc.)
Breed		Breed of the pet
AgeMonths		Age in months
Color		Color of the pet
WeightKg		Weight in kilograms
Status		Adoption status (Available, Adopted, etc.)

3. Application

Attribute	Key	Description
ApplicationID	PK	Unique ID for each application
<i>ApplicantID</i>	FK → Applicant.ApplicantID	Applicant who submitted the application
<i>PetID</i>	FK → Pet.PetID	Pet being applied for
SubmissionDate		Date the application was submitted
Status		Application stage (Pending, Accepted, Denied, Completed)

4. Adoption

Attribute	Key	Description
AdoptionID	PK	Unique adoption record
<i>PetID</i>	FK → Pet.PetID (UNIQUE)	Adopted pet
<i>ApplicantID</i>	FK → Applicant.ApplicantID	Adopter
<i>ApplicationID</i>	FK → Application.ApplicationID	Originating approved application
AdoptionDate		Date adoption finalized

5. MedicalRecord

Attribute	Key	Description
RecordID (or VisitID)	PK	Unique record for the visit
<i>PetID</i>	FK → Pet.PetID	Pet receiving treatment
VisitDate		Date of medical exam or vaccination
RecordType		Exam, Vaccine, Surgery, etc.
Diagnosis		Notes on diagnosis
Treatment		Treatment administered
VaccineType		Vaccine name (if applicable)

Entity Relationships

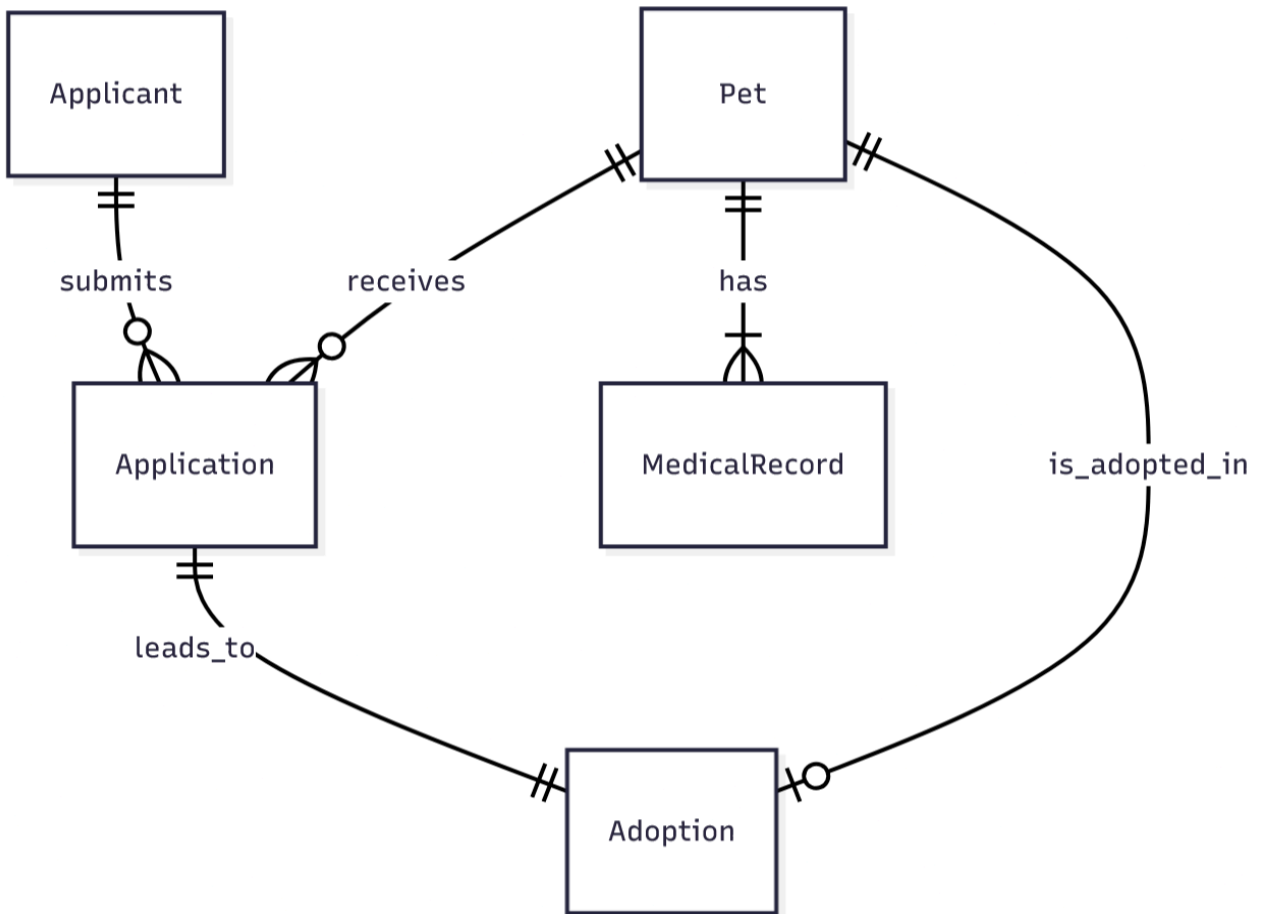
Relationships Summary

Relationship	Type	Description
Applicant → Application	1-to-many (1:N)	One applicant can submit many applications
Pet → Application	1-to-many (1:N)	One pet can receive many applications
Application → Adoption	1-to-(0 or 1)	An application can lead to at most one adoption
Pet → Adoption	1-to-(0 or 1)	Each pet can have at most one adoption
Pet → MedicalRecord	1-to-many	One pet has one or many medical records

Diagrams and Designs

Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) provides a visual representation of the database entities and the relationships between them.



Relational Schema

The relational schema translates the conceptual ERD into a relational database structure, identifying primary keys, foreign keys, and the relationships between the resulting tables.

